

Table 1: Experimental Performance of MaACO vs Baselines with Pairwise Wilcoxon Test Signs (+, −, ≈ at $\alpha = 0.05$).

Problem	M	NSGA3		MaACO	
		HV	Δ_p	HV	Δ_p
ZDT1	2	0.8669 ≈	0.0068 ≈	0.8715	0.0038
ZDT2	2	0.5353 ≈	0.0061 ≈	0.5384	0.0038
ZDT3	2	1.3088 ≈	0.0187 ≈	1.3092	0.0177

Table 2: Non-Parametric Statistical Analysis: Friedman Average Ranks and Holm Post-Hoc Procedure (Control: MaACO, $\alpha = 0.05$).

Algorithm	Friedman Rank (HV)	Friedman Rank (Δ_p)	z -statistic	Raw p -value	Holm p_{adj}	$\mathbb{I}$
NSGA3	2.00	2.00	-1.732	0.0833	0.0833	F
MaACO (Control)	1.00	1.00	—	—	—	—
Friedman Test (HV): $\chi_F^2 = 3.000$, $p = 0.0833$						